

Welcome to Stat153 / 248

Introduction to Time Series Analysis

- We will start at 10 minutes after the hour (Berkeley time)
- While you wait, you can fill out the pre-semester survey:
- <https://forms.gle/2jLWHGeAvXe9EMcG9> (or use QR code) - use berkeley.edu email to access



Lecture 1: Introduction to the Course

Stat 153/248 - Introduction to Time Series Analysis

**Liberty Hamilton
Spring 2026**

Agenda

- Introductions
- Structure of the course
- What is a time series?
- Overview of what you'll learn

Introductions

- **Instructor:** Liberty Hamilton (Statistics and Neuroscience Departments) - liberty.hamilton@berkeley.edu
- **GSI:** Nicholas Liskij, Yichen Pan
- Office hours starting next week

Structure of the course

- Course website: stat153.berkeley.edu/spring-2026
- Be sure to look at the syllabus
- Lecture materials (slides, notes, code) will be posted *after* each lecture
- Materials for labs and homework will be posted
- Links to bCourses, Ed Discussion, etc.
- Related reading (PDFs, links)

Assignments

- 5 homework assignments (Stat 248 includes some additional problems)
- 1 midterm (Stat 153 and Stat 248)
- Final exam (Stat 153) or final project (Stat 248)
- Projected dates on syllabus

Prerequisites

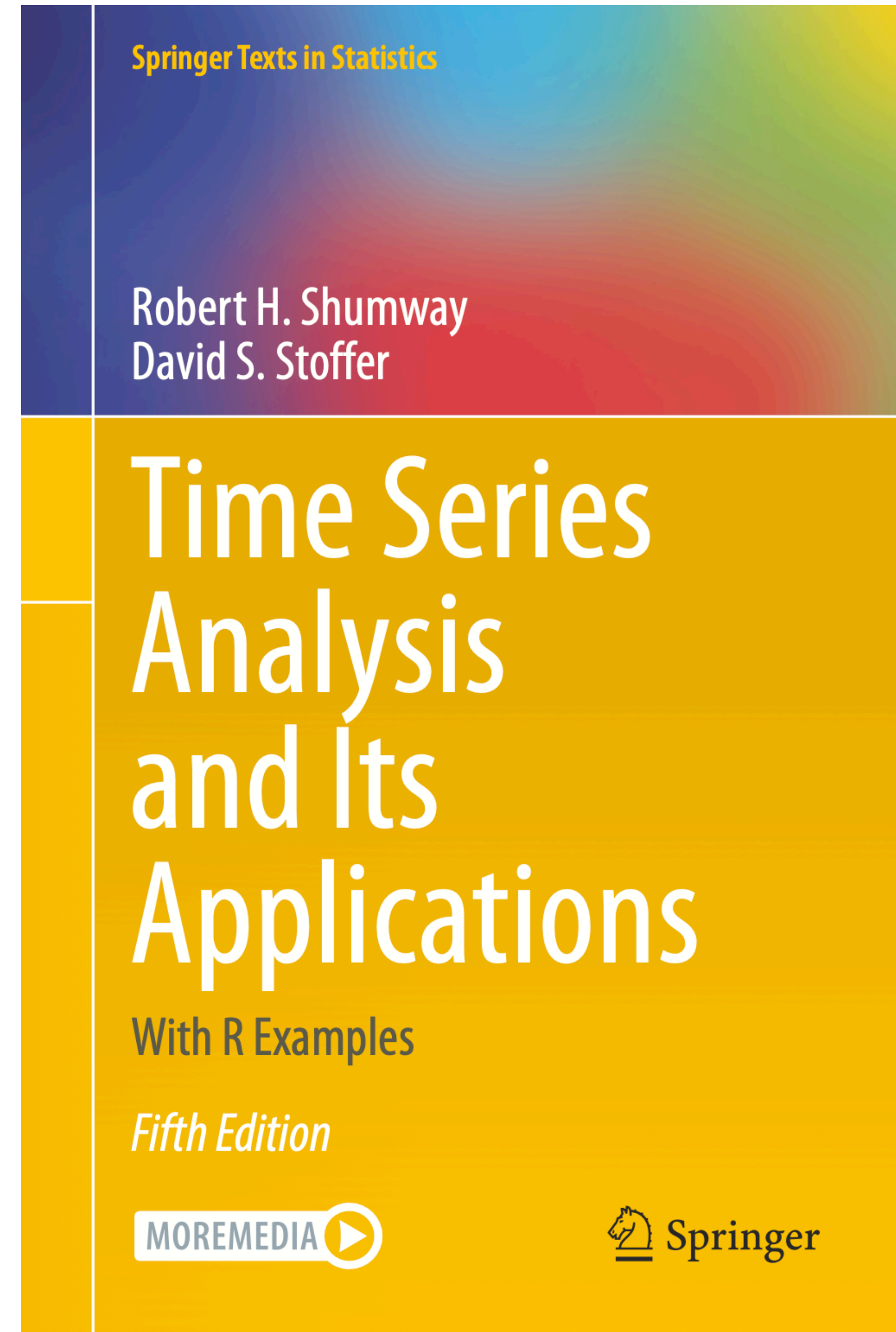
- Probability at the level of Stat 134 or Data 140 or EE 126 required
- 🐍 We assume basic fluency in Python
 - Many assignments will take advantage of the [Berkeley datahub](#), but for others you should have Python installed on your computer (I recommend the [Anaconda python](#) distribution).
 - Labs will use JupyterLab/Jupyter notebooks

Introduce yourselves!

- 🖐️ Introduce yourself to 2-3 people sitting near you
- 🎓 Are you a grad student or undergrad?
- ⓘ What's your name, major, interests outside of this class?

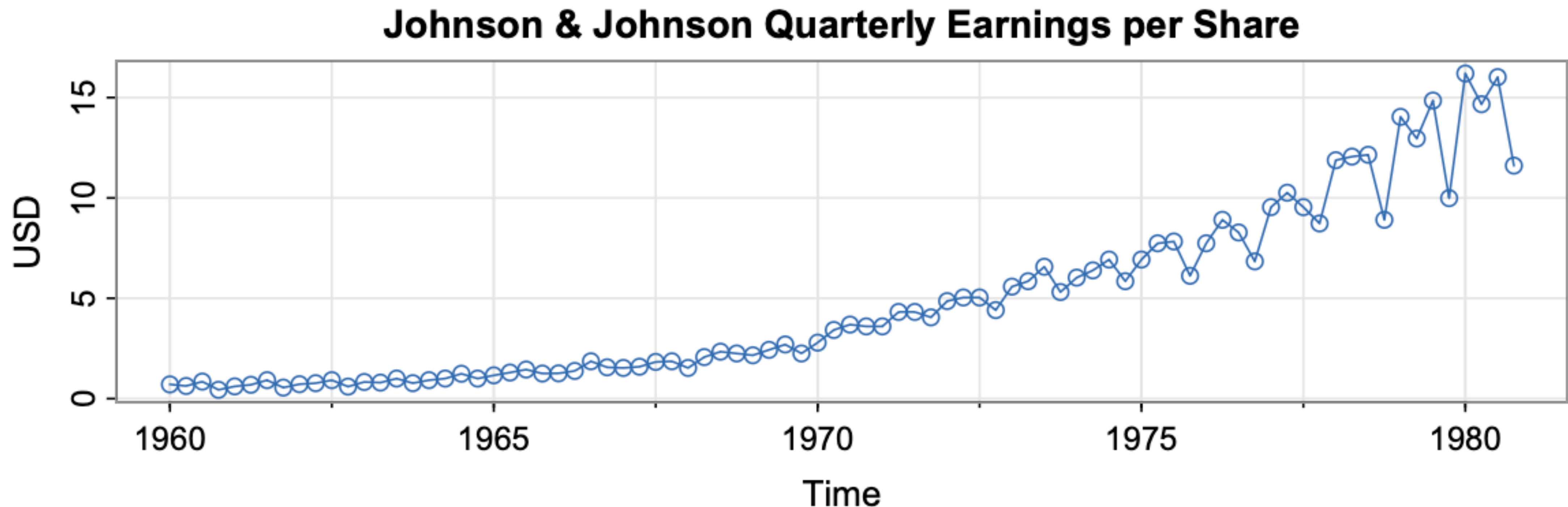
Let's start!

- Related reading for today and next week: Chapter 1 of Shumway and Stoffer (SS)
- Available **free** online (see course website for a link)
- We'll use this book towards the beginning of the semester



What is a time series?

- A time series is a sequence of data points collected over time
- Examples: hourly temperature readings, stock market fluctuations, brain responses over time to a stimulus



Why should we care about time series?

- We can uncover regular patterns over time
- Includes cycles or trends

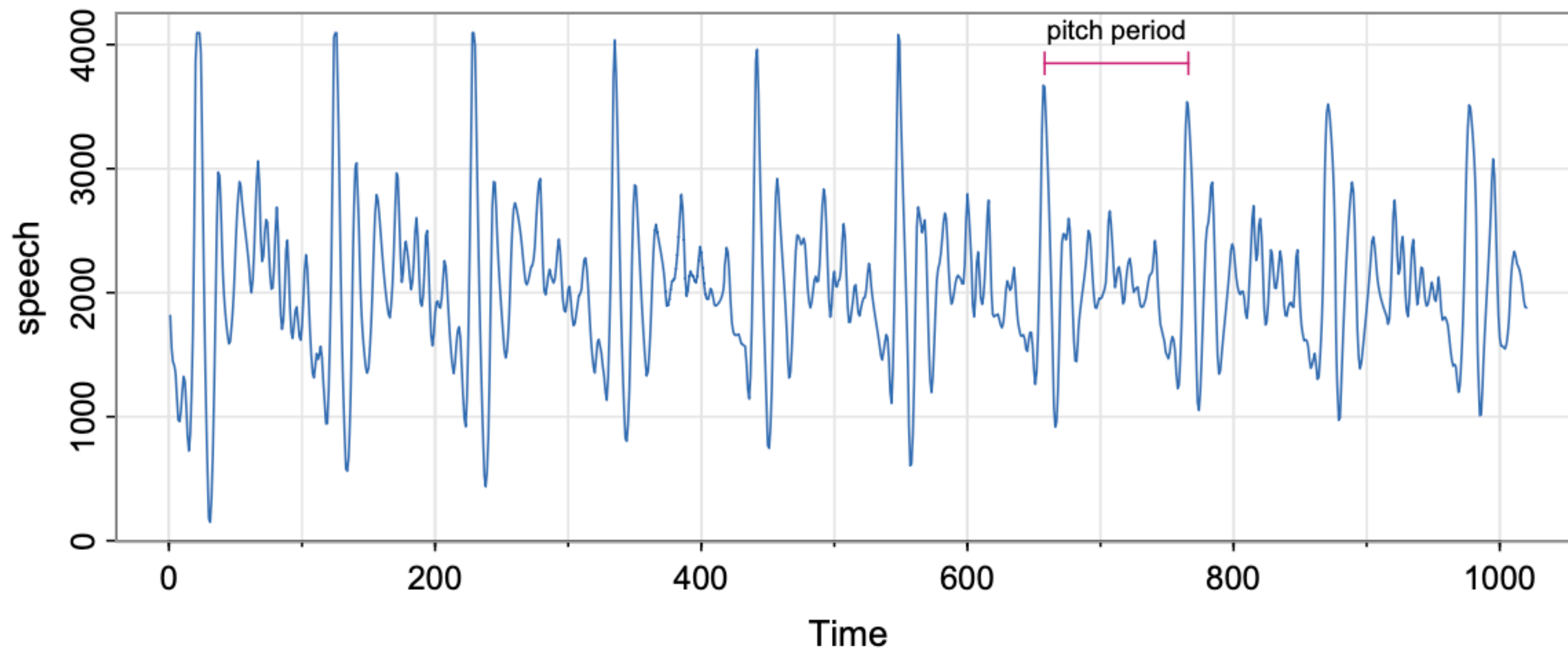


Fig. 1.3. Speech recording of the syllable *aaahhh* sampled at 10,000 points per second with $n = 1020$ points

Where might we see time series data?

- Neuroscience
- Medicine
- Economics
- Epidemiology
- Social science
- Astronomy
- Large language models (LLMs) ...and more!

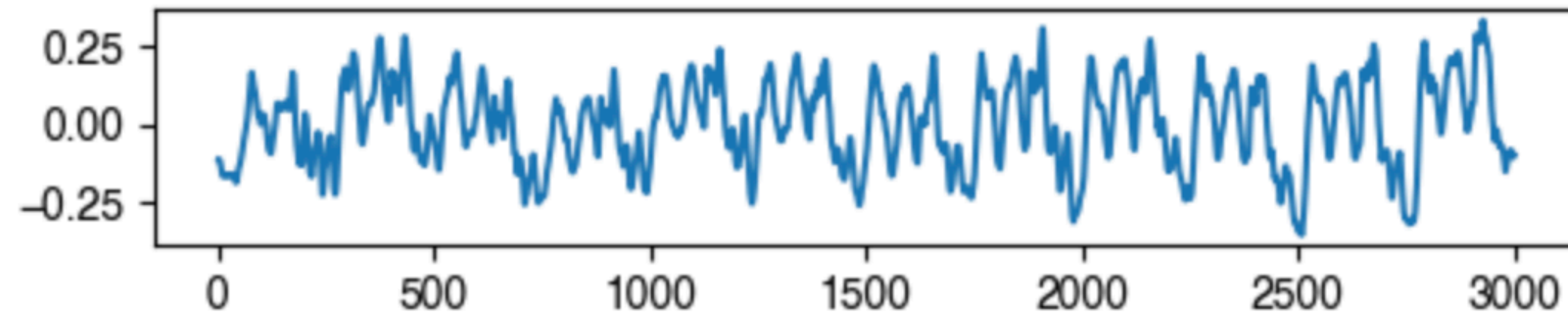
What is and is not a time series?

- Go to poll: <https://pollev.com/stat153>

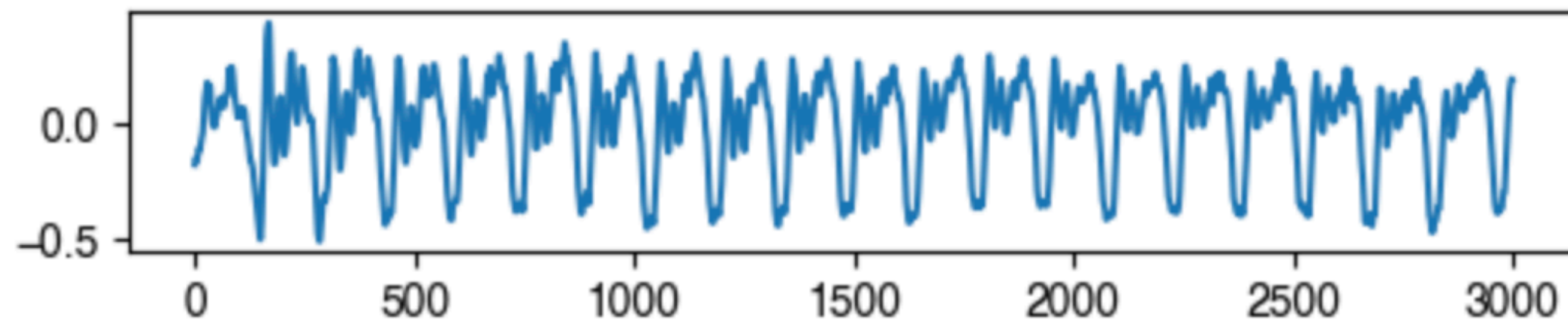


A real example (sounds)

Sound A: low "uh" /ə/



Sound B: high "uh" /ə/



What is different about time series data?

- Not i.i.d. (independent and identically distributed)
- Time dependence / correlation structure
- Can have drift over time
- Let's step through some examples and discuss...

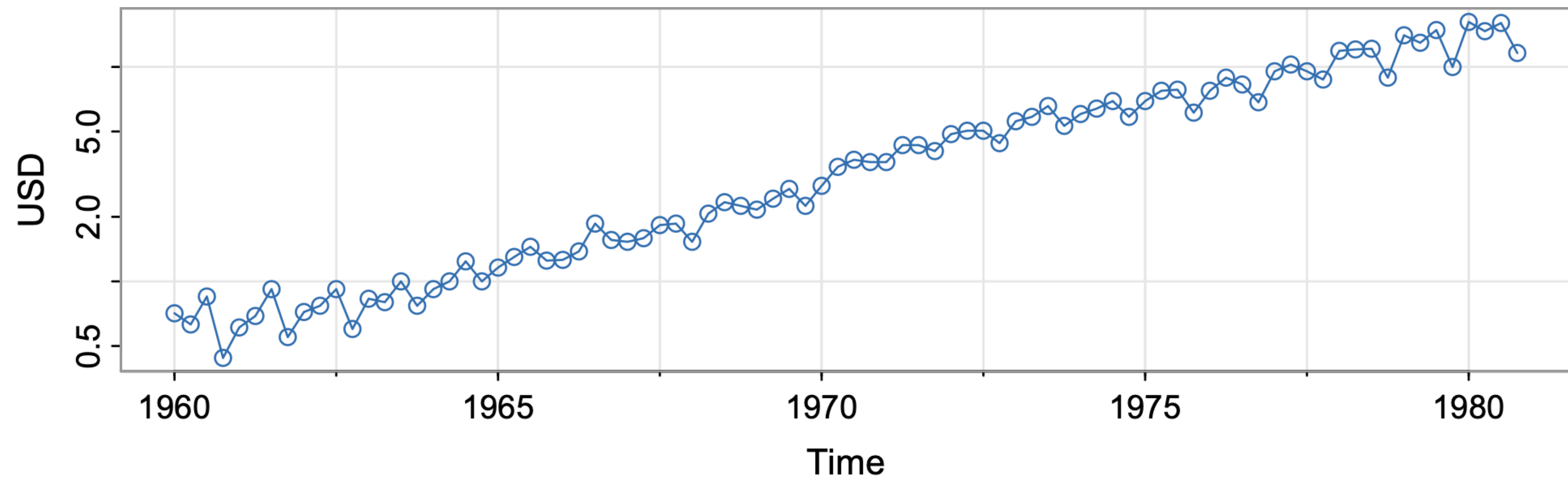
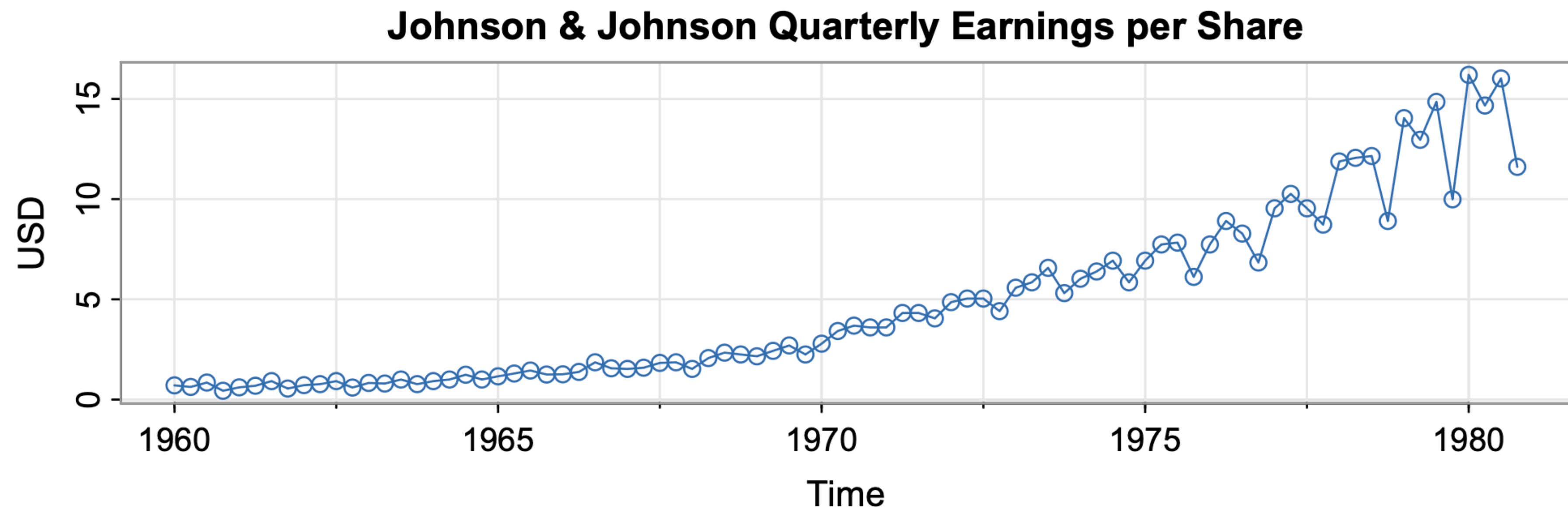


Fig. 1.1. Johnson & Johnson quarterly earnings per share in US dollars, 1960-I to 1980-IV (top). The same data on a log scale (bottom)

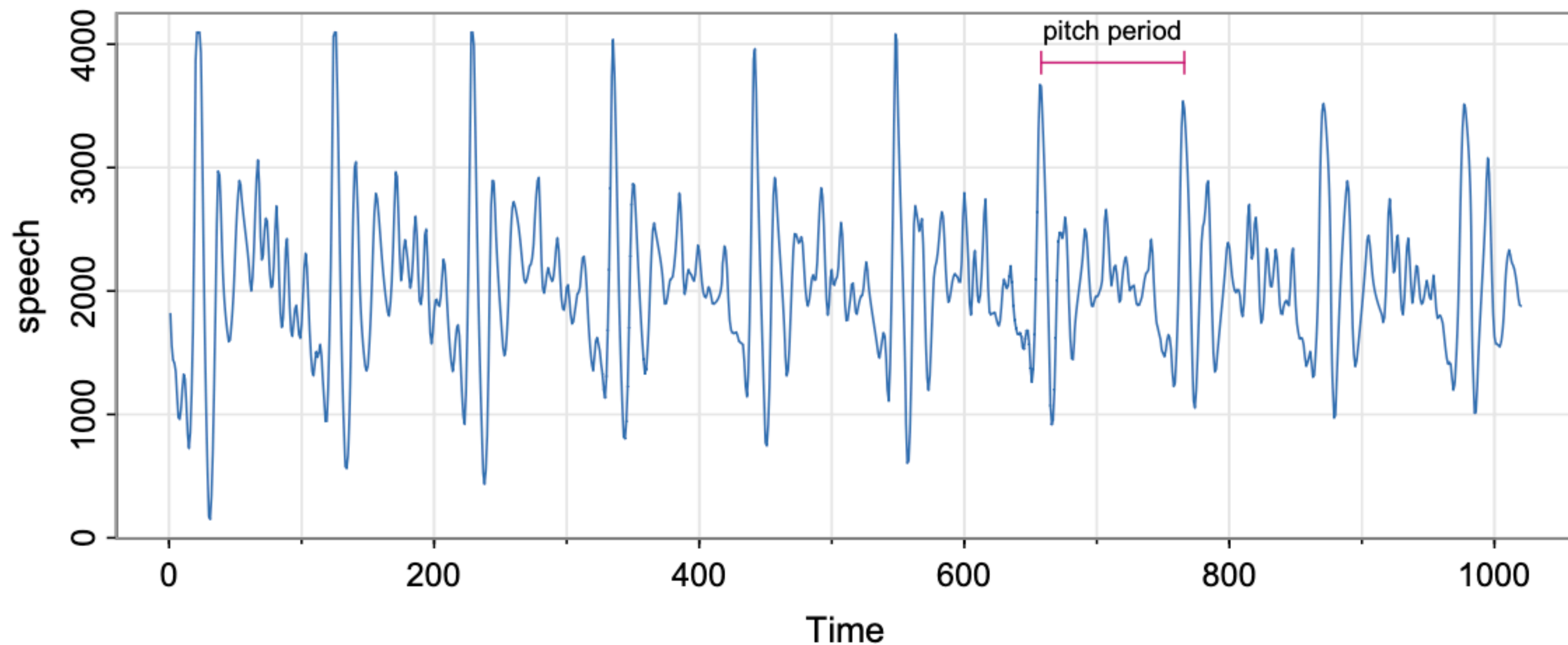


Fig. 1.3. Speech recording of the syllable *aaahhh* sampled at 10,000 points per second with $n = 1020$ points

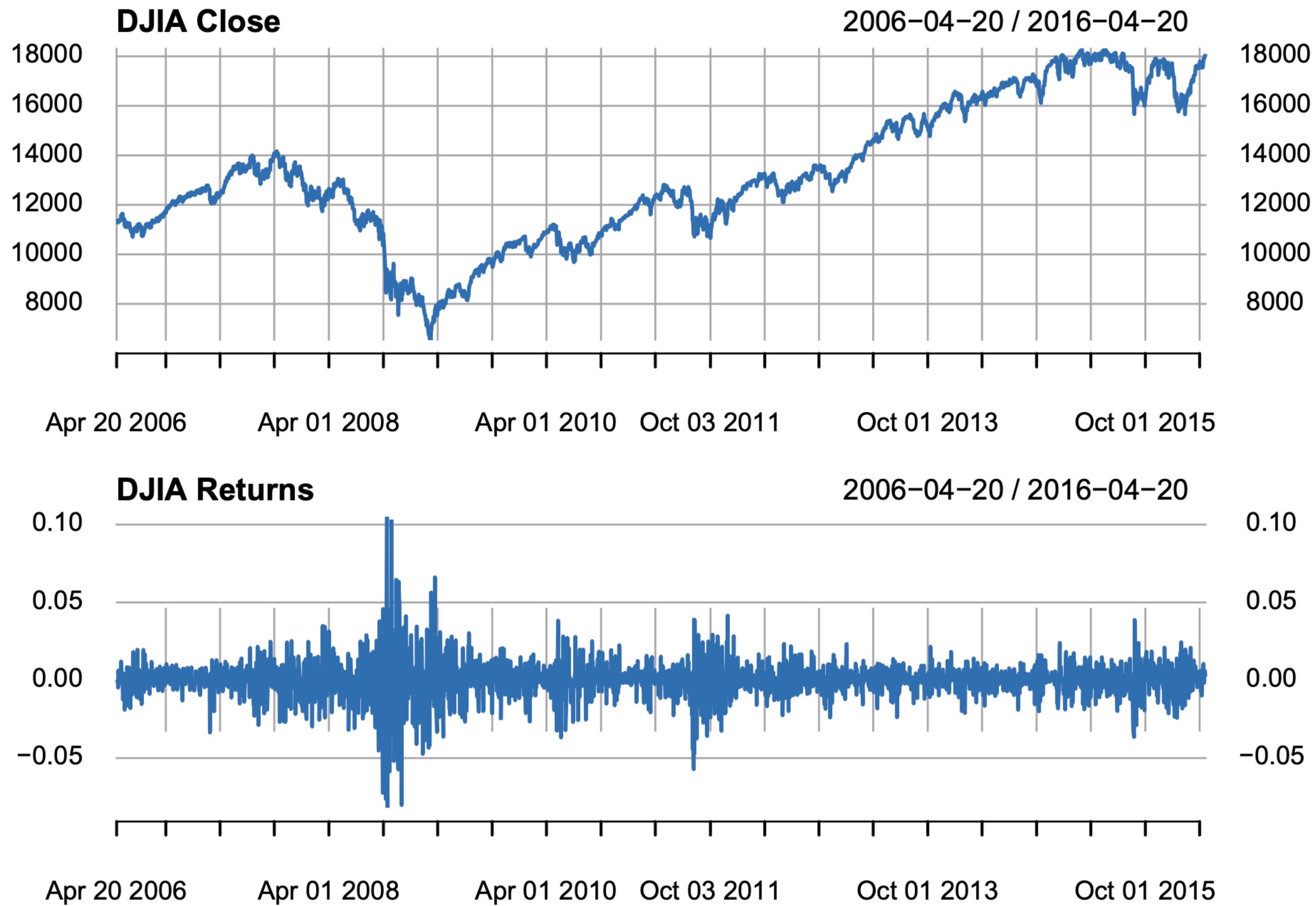


Fig. 1.4. Dow Jones Industrial Average (DJIA) trading days closings (top) and returns (bottom) from April 20, 2006 to April 20, 2016

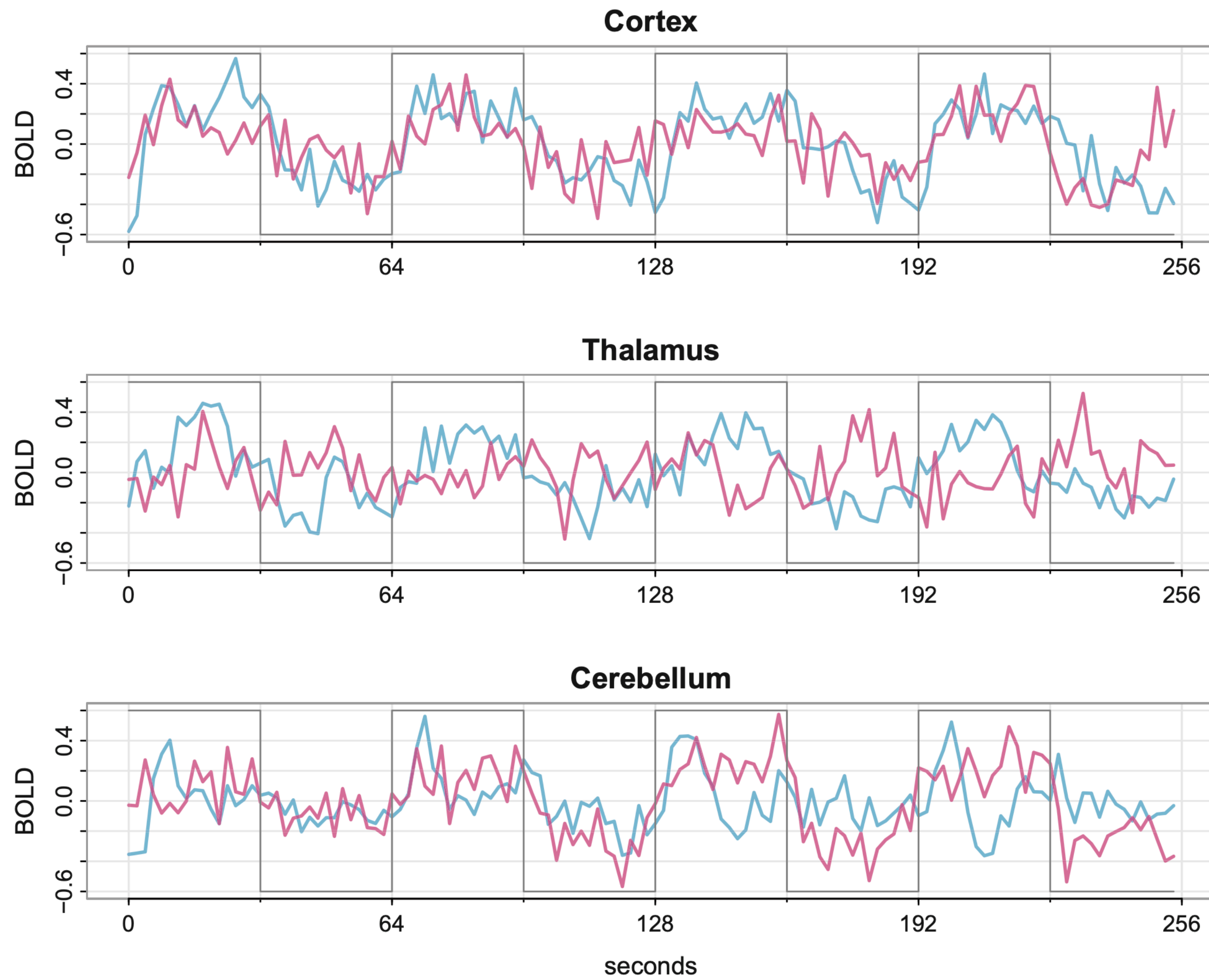
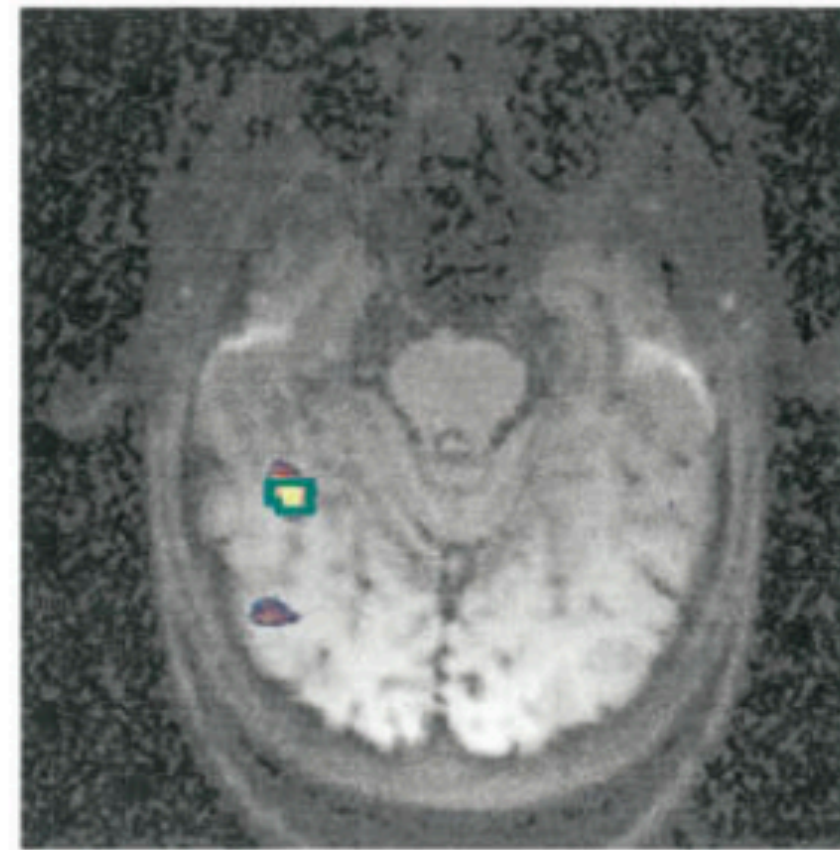
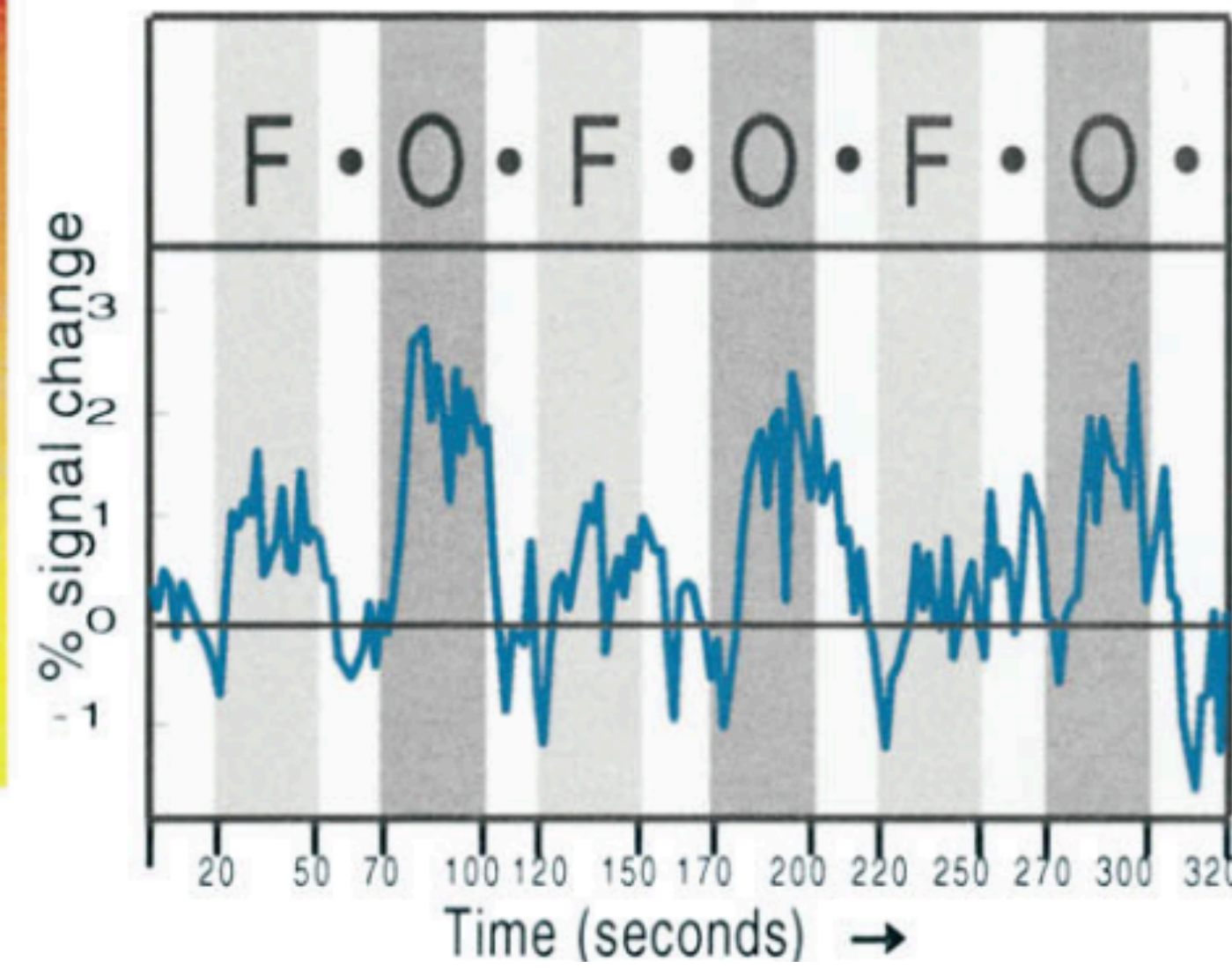
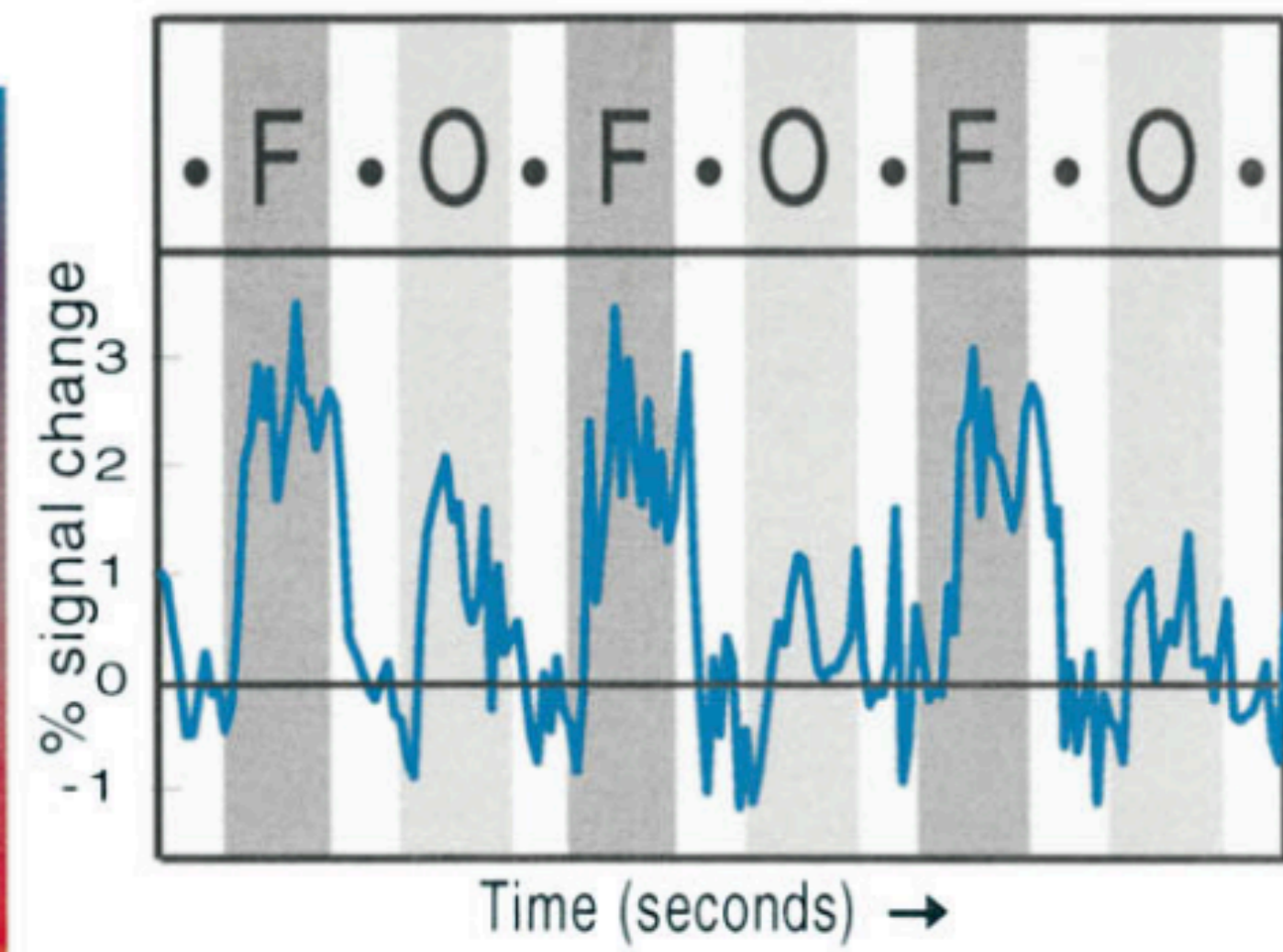
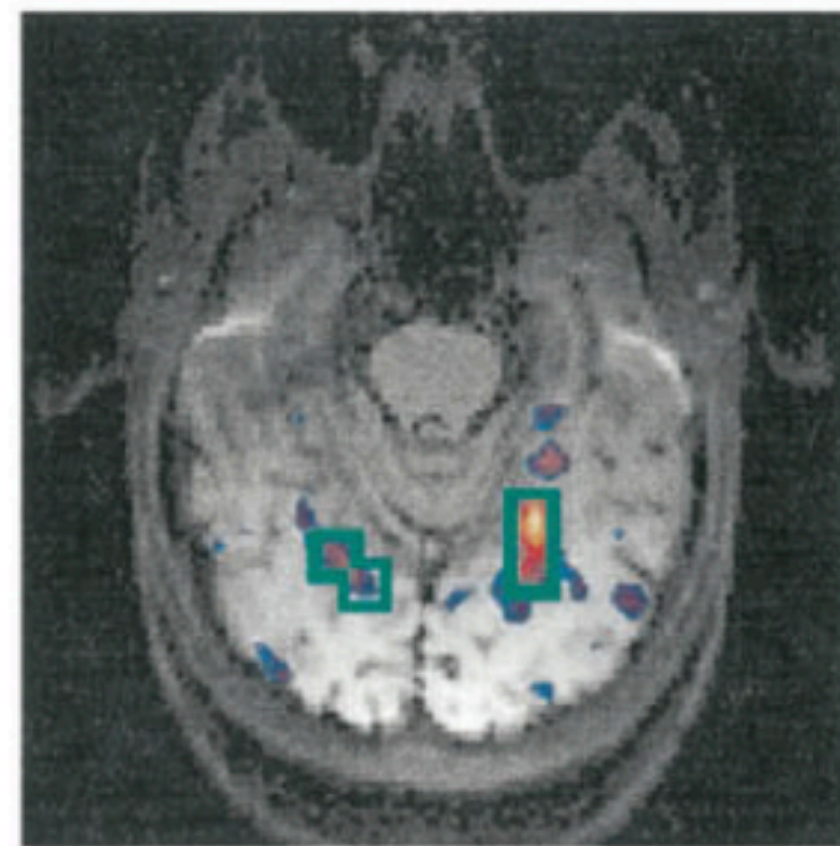


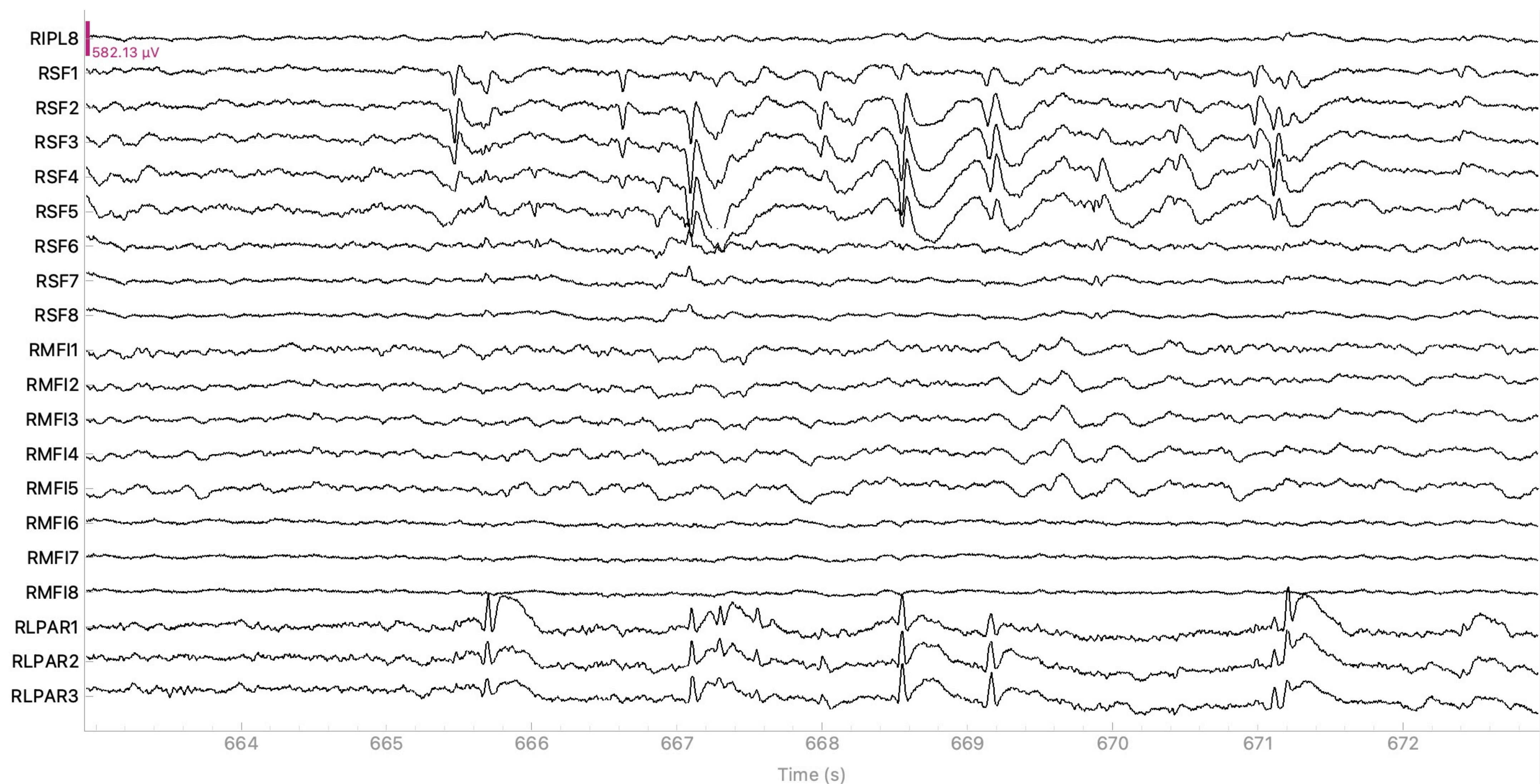
Fig. 1.7. fMRI data from various locations in the cortex, thalamus, and cerebellum; $n = 128$ points, one observation taken every 2 seconds. The square wave shows the stimulus signal (on or off)

1a. Faces > Objects

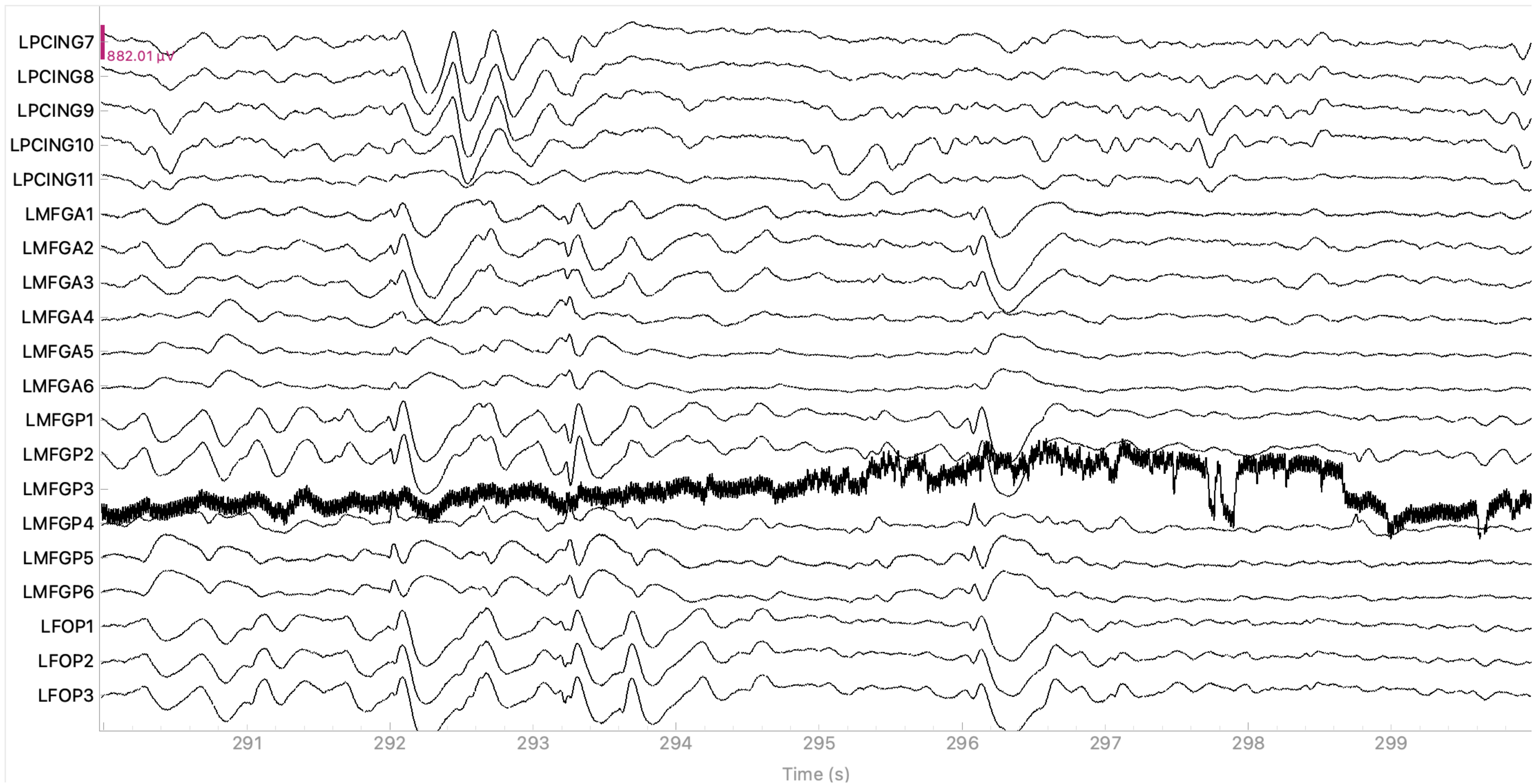


1b. Objects > Faces





Example brain recordings from various locations in the frontal and parietal lobes in a patient with epilepsy undergoing surgical monitoring. Recorded at 512 Hz sampling rate, one observation every ~2 milliseconds. (Hamilton lab, unpublished data)



Course topics

- Measures of dependence
- Linear regression - simple, multiple, nonlinear, regularization
- Power spectral analysis and time-frequency analysis
- AR models
- ARIMA models
- Time-lagged regression
- State-space models
- Neural networks - CNNs, RNNs, self-supervised learning (WavLM)

U.S. Population

<https://fred.stlouisfed.org/series/POPTHM>

Observations 

Sep 2025: **342,741**

Updated: Dec 5, 2025 9:04 AM CST

Next Release Date: Jan 29, 2026

Units:

Thousands,
Not Seasonally Adjusted

Frequency:

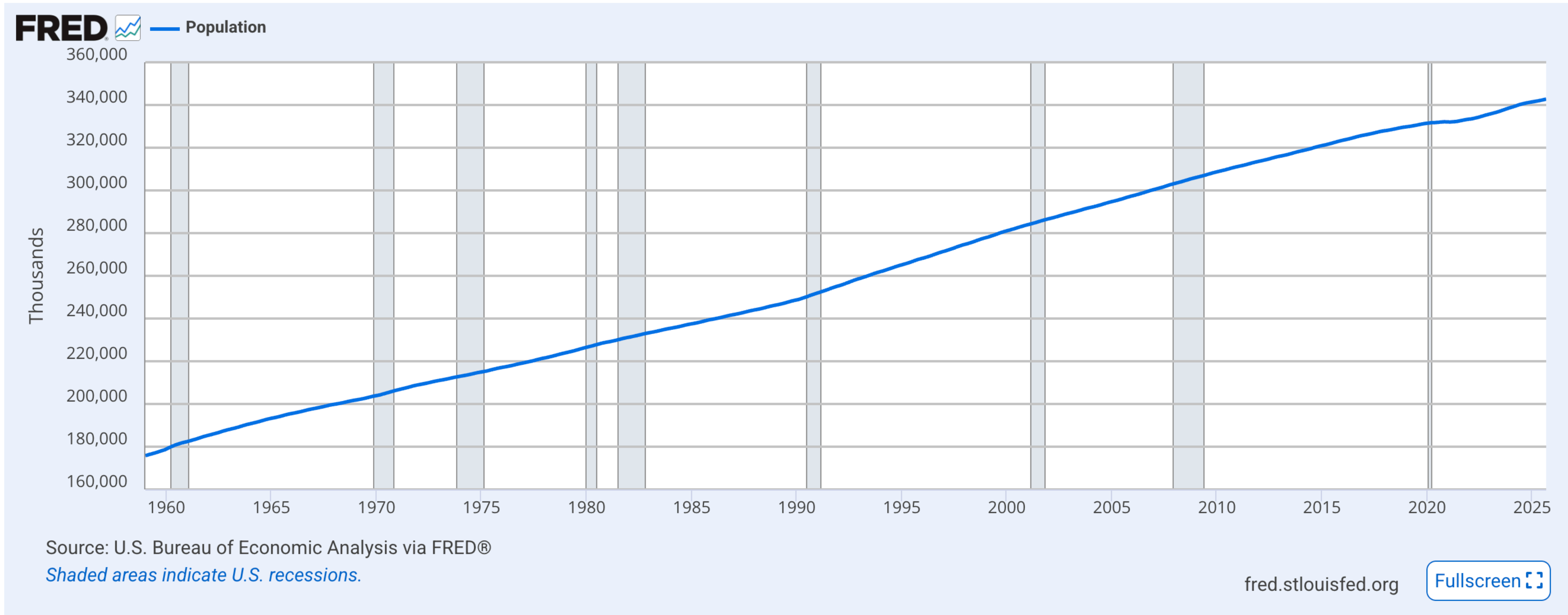
Monthly

1Y | 5Y | 10Y | Max

1959-01-01 to 2025-09-01

Edit Graph 

Download 



Questions

- What will the U.S. population be in 2050?
- What is the rate of growth of the population?
- Has the growth rate been constant over time?
- In what periods has population been fastest / slowest?

Time Series Analysis and Prediction

One of the main questions we will address in this class is how to **predict future values based on current data**

Simple linear regression

```
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
df=pd.read_csv('POPTHM.csv')
df
```

	observation_date	POPTHM
0	1959-01-01	175818
1	1959-02-01	176044
2	1959-03-01	176274
3	1959-04-01	176503
4	1959-05-01	176723
...	...	...
796	2025-05-01	342032
797	2025-06-01	342197
798	2025-07-01	342370
799	2025-08-01	342555
800	2025-09-01	342741

801 rows x 2 columns

For a simple linear regression, we fit the relationship between our output y_t and time:

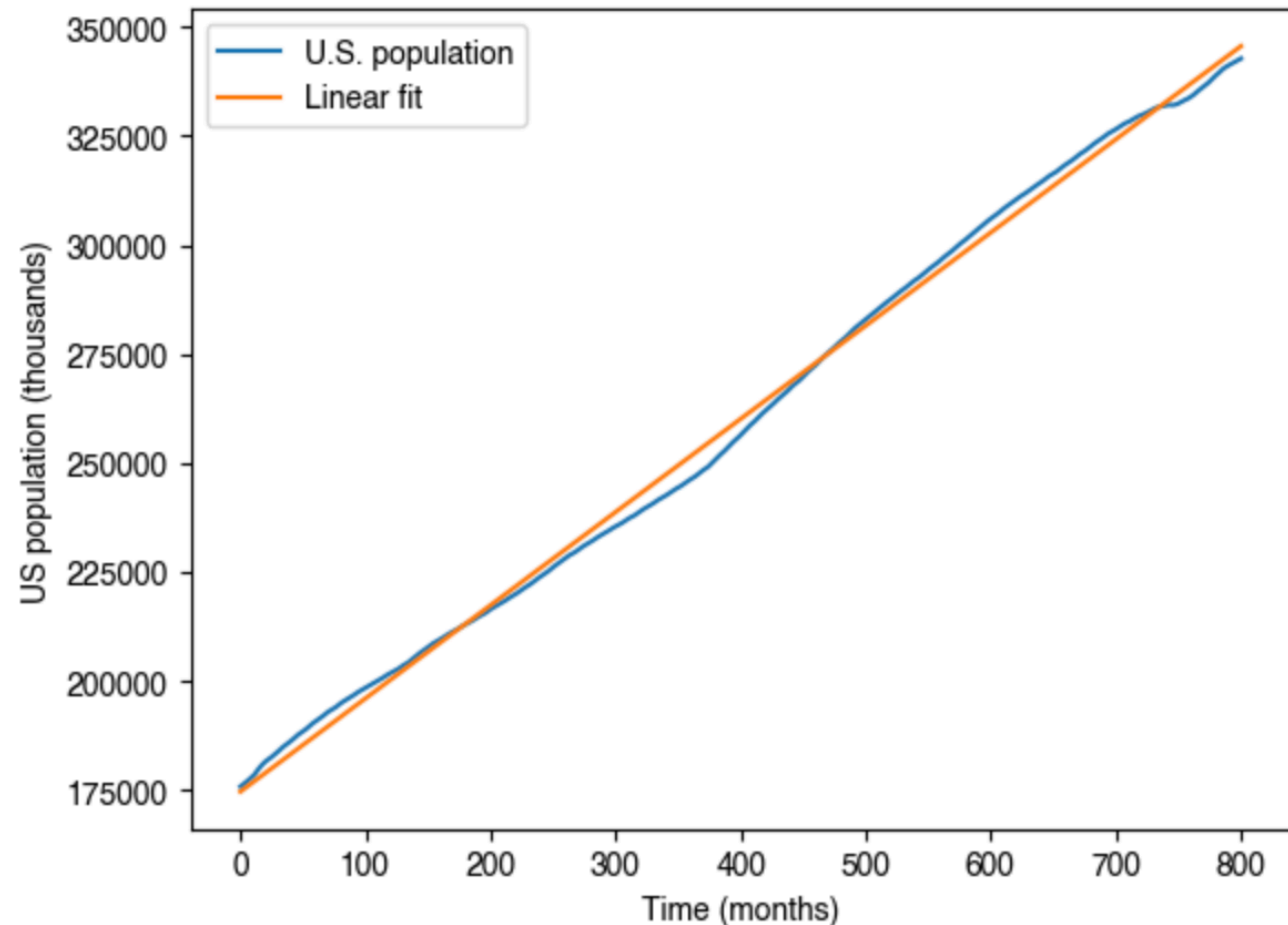
$$y_t = \beta_0 + \beta_1 x_t$$

Simple linear regression - US population

```
x = np.arange(len(df['POPTHM']))
y = df['POPTHM']

# Fit a line  $y = mx + b$ 
m, b = np.polyfit(x, y, 1)

# Plot data and fit
plt.plot(x, y, label='U.S. population')
plt.plot(x, m*x+b, label='Linear fit')
plt.xlabel('Time (months)')
plt.ylabel('US population (thousands)')
plt.legend()
```



What if we look at population change over time?

Observations 

Sep 2025: **186**

Updated: Dec 5, 2025 9:04 AM CST

Next Release Date: Jan 22, 2026

Units:

Change, Thousands,
Not Seasonally Adjusted

Frequency:

Monthly

1Y

5Y

10Y

Max

1959-02-01

to

2025-09-01

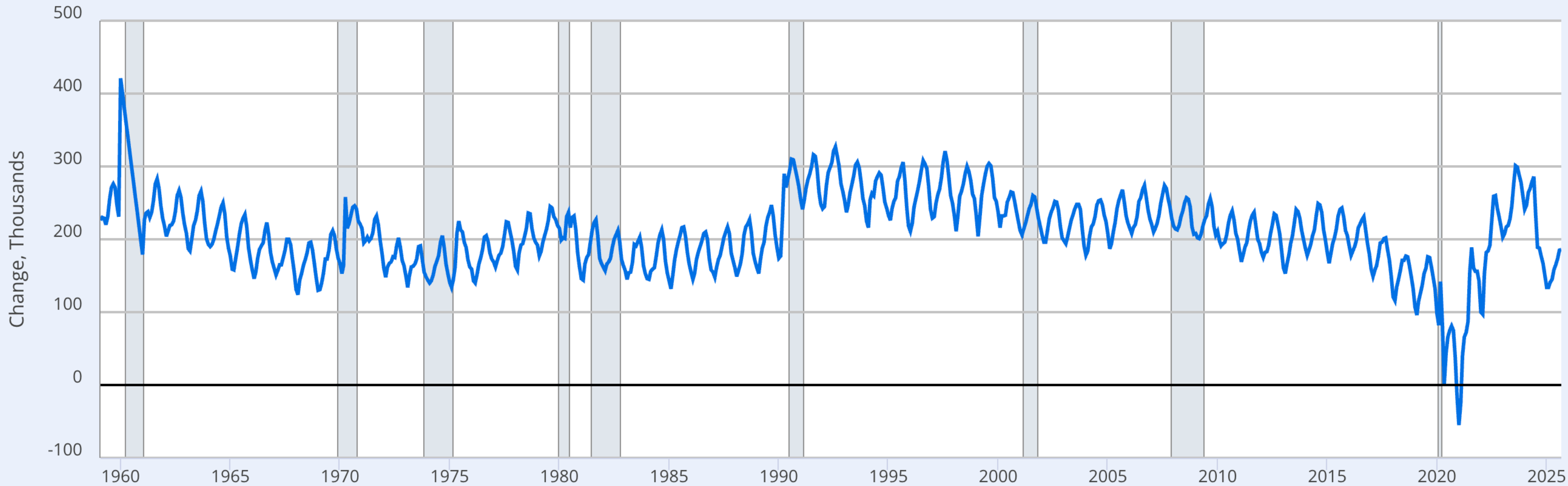
Edit Graph 

Download 

FRED



Population



Source: U.S. Bureau of Economic Analysis via FRED®

Shaded areas indicate U.S. recessions.

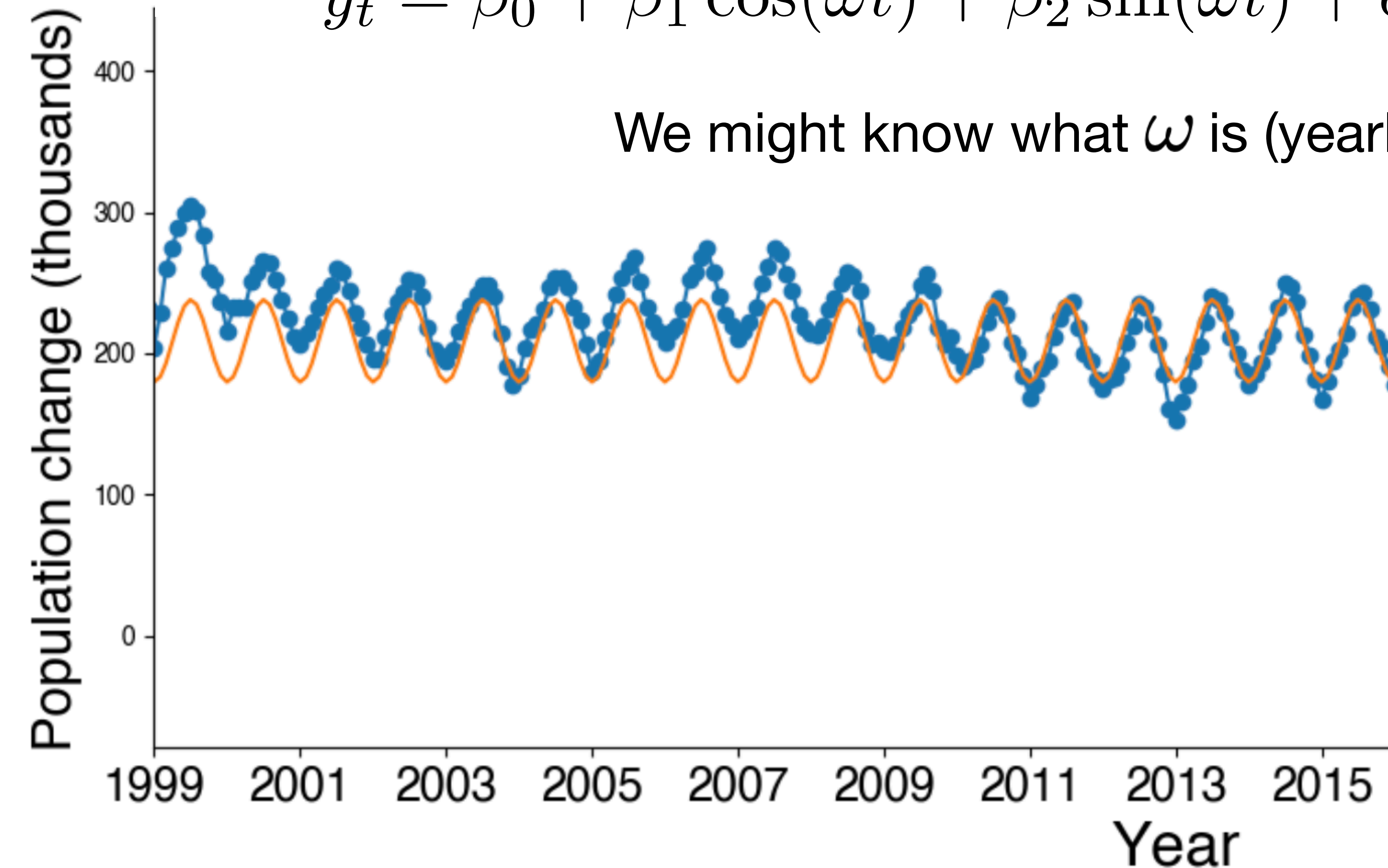
fred.stlouisfed.org

Fullscreen 

Seasonal trends - sinusoidal regression

$$y_t = \beta_0 + \beta_1 \cos(\omega t) + \beta_2 \sin(\omega t) + \epsilon$$

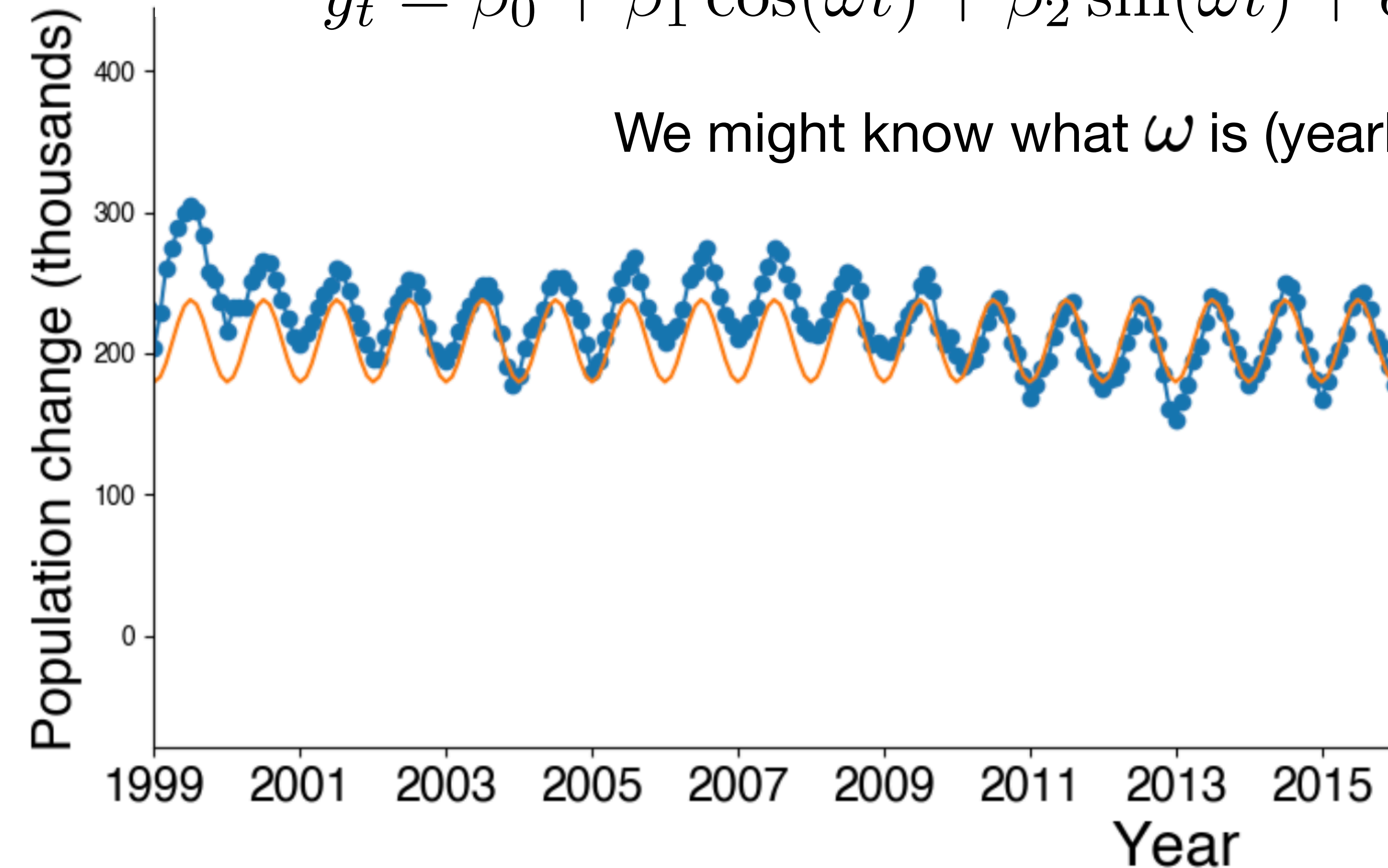
We might know what ω is (yearly trend?)



Seasonal trends - sinusoidal regression

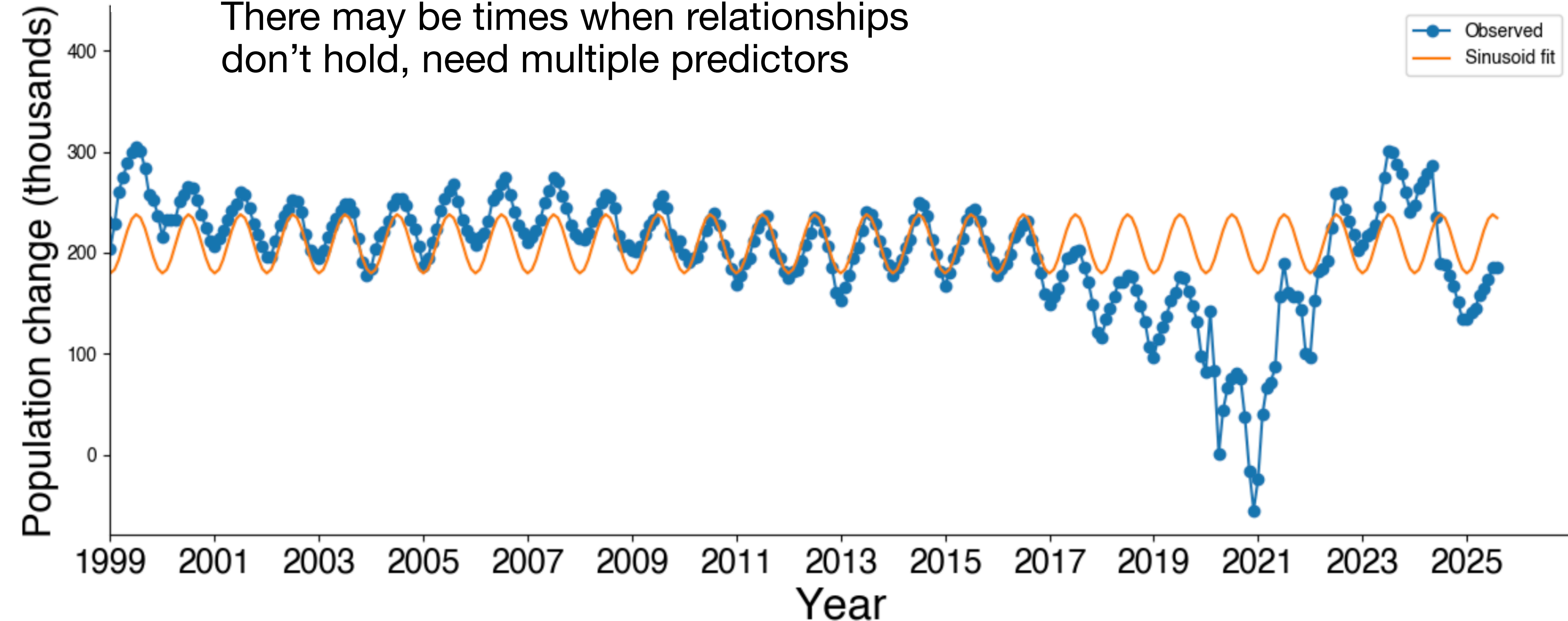
$$y_t = \beta_0 + \beta_1 \cos(\omega t) + \beta_2 \sin(\omega t) + \epsilon$$

We might know what ω is (yearly trend?), or we can fit it

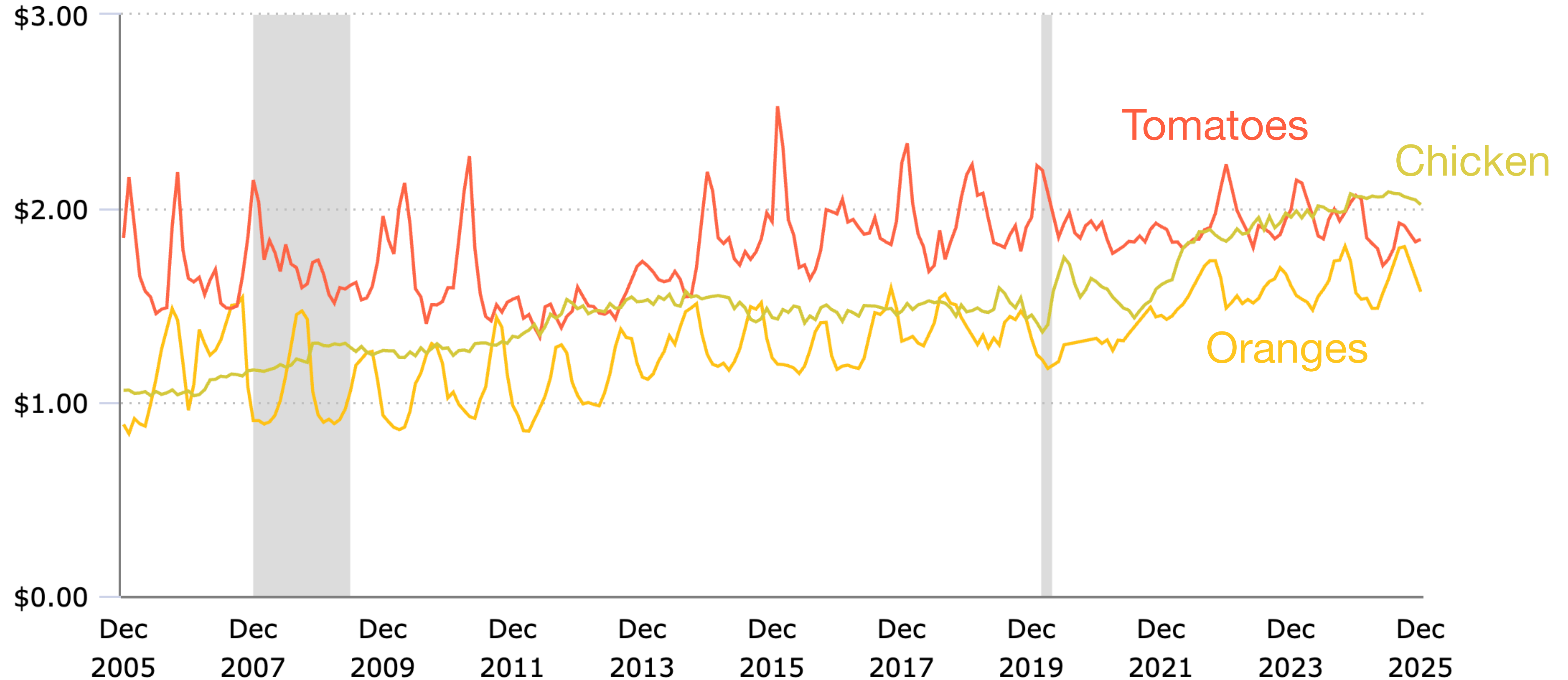


Seasonal trends - sinusoidal regression

There may be times when relationships don't hold, need multiple predictors



Average price per pound of various goods



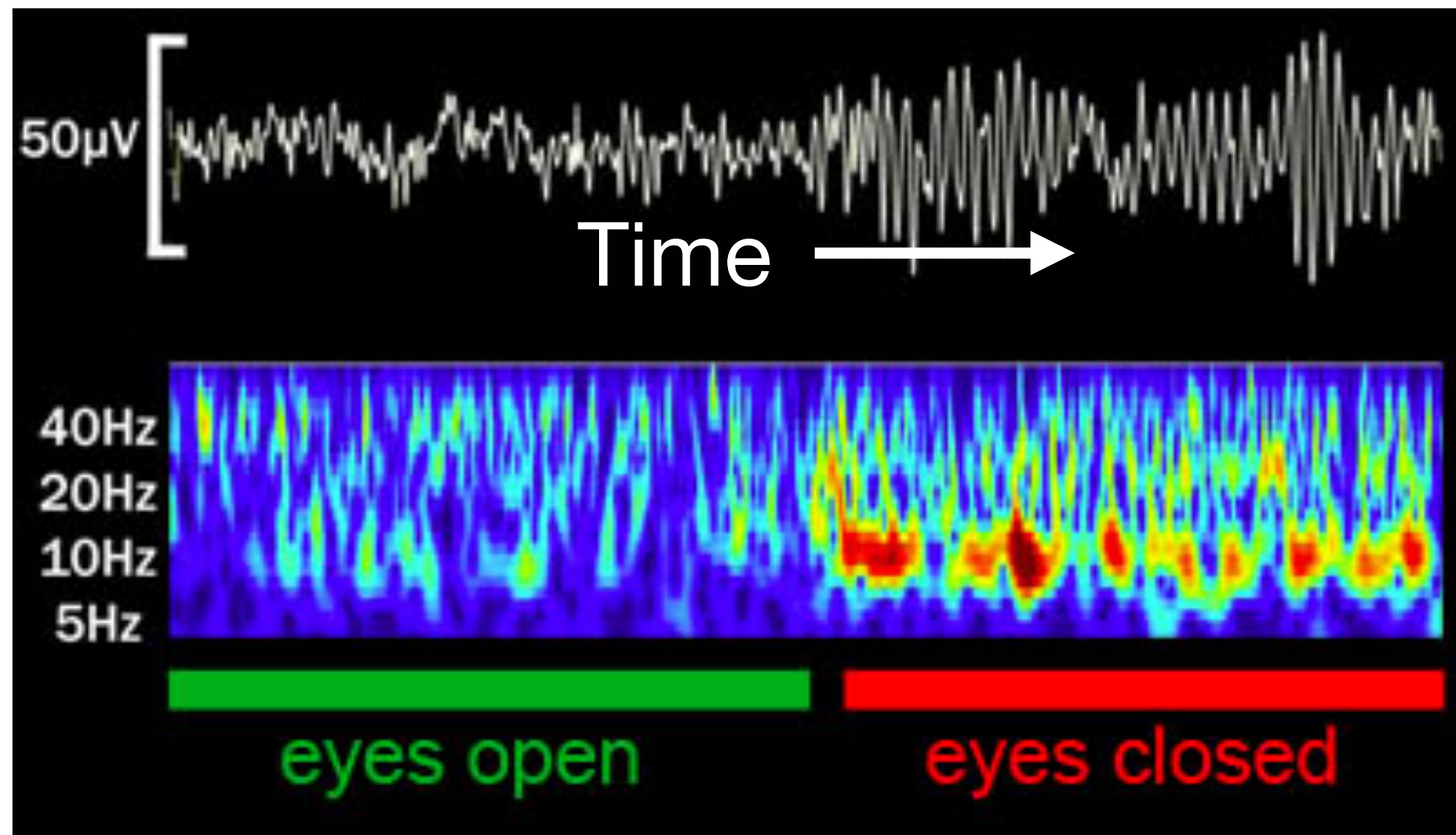
<https://www.bls.gov/charts/consumer-price-index/consumer-price-index-average-price-data.htm>

Sinusoidal models and Fourier Analysis

- $y_t = \beta_0 + \beta_1 \cos(\omega t) + \beta_2 \sin(\omega t) + \epsilon$
- Models like this are closely related to **Fourier Analysis**, which describes complex, periodic signals as a sum of simpler waveforms
- These models are used extensively in engineering, physics, neuroscience, audio signal processing

Time-frequency analysis

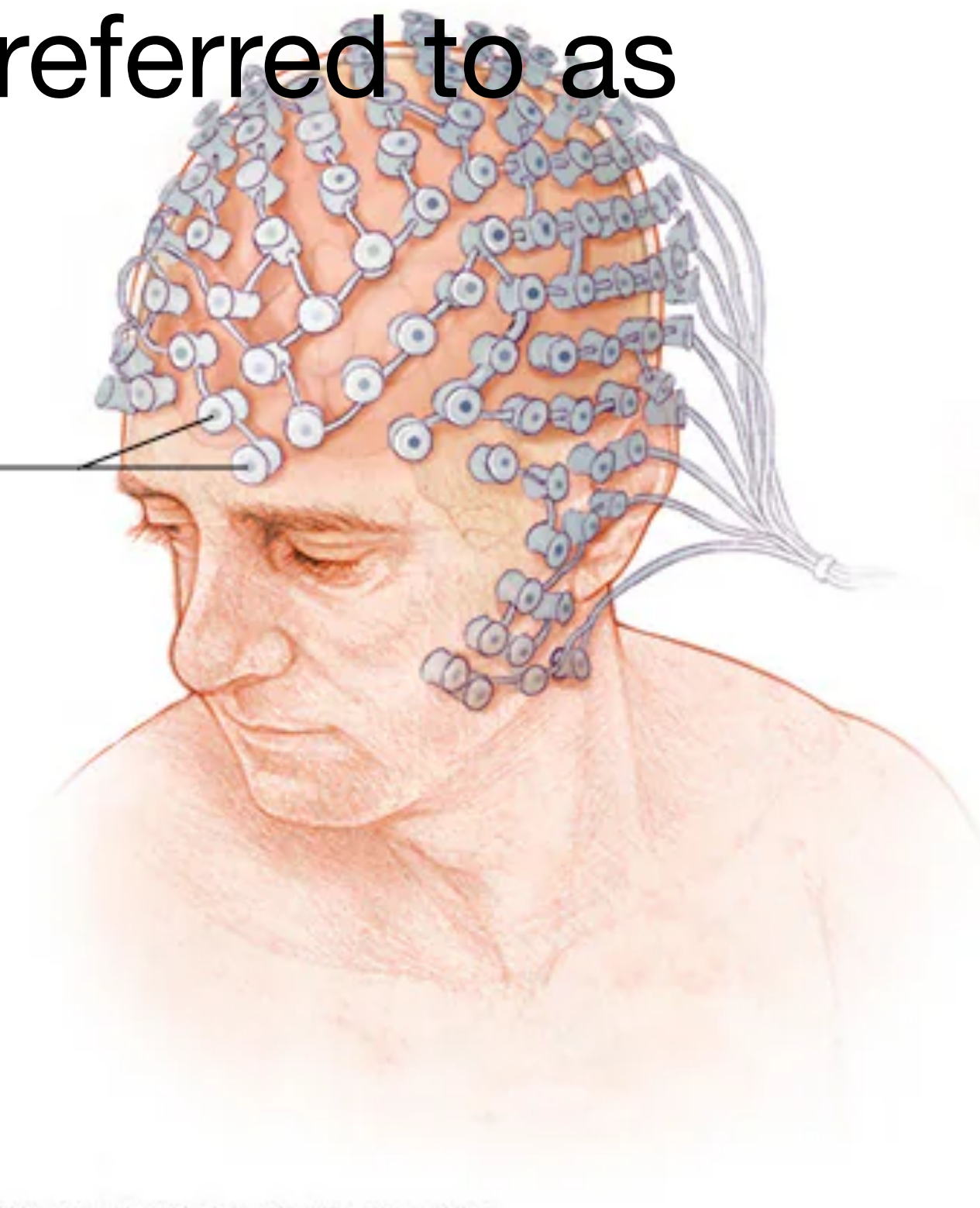
- Example below shows the EEG time series recording and time-frequency spectrogram from an electrode on the back of the head with eyes open vs. closed
- The large amplitude waves during “eyes closed” are referred to as the “alpha rhythm”

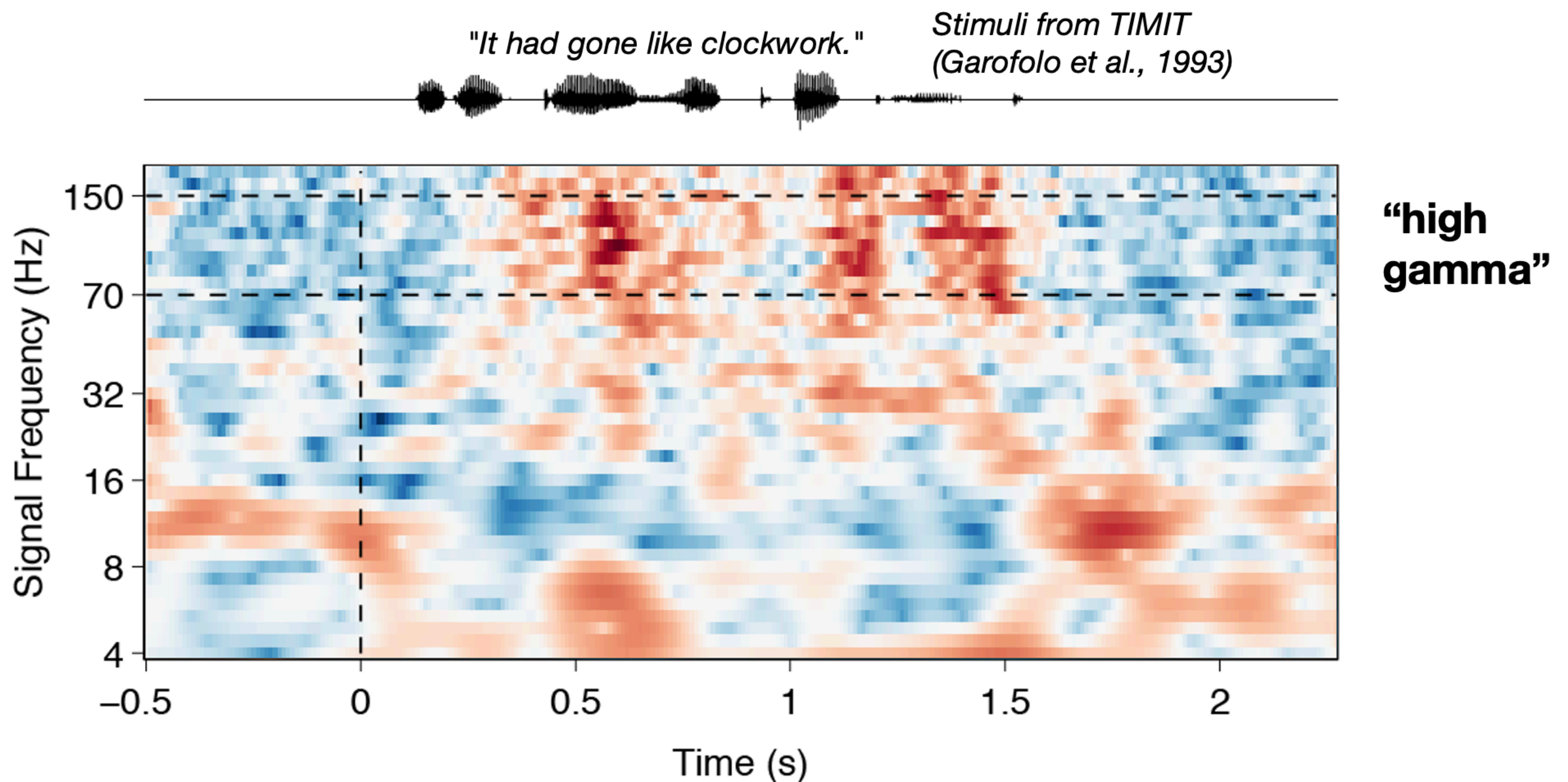


waveform

EEG electrodes

spectrogram





- For some types of neural data, specific frequency bands may better relate to sensory input - e.g. “high gamma” frequencies reflect responses to speech

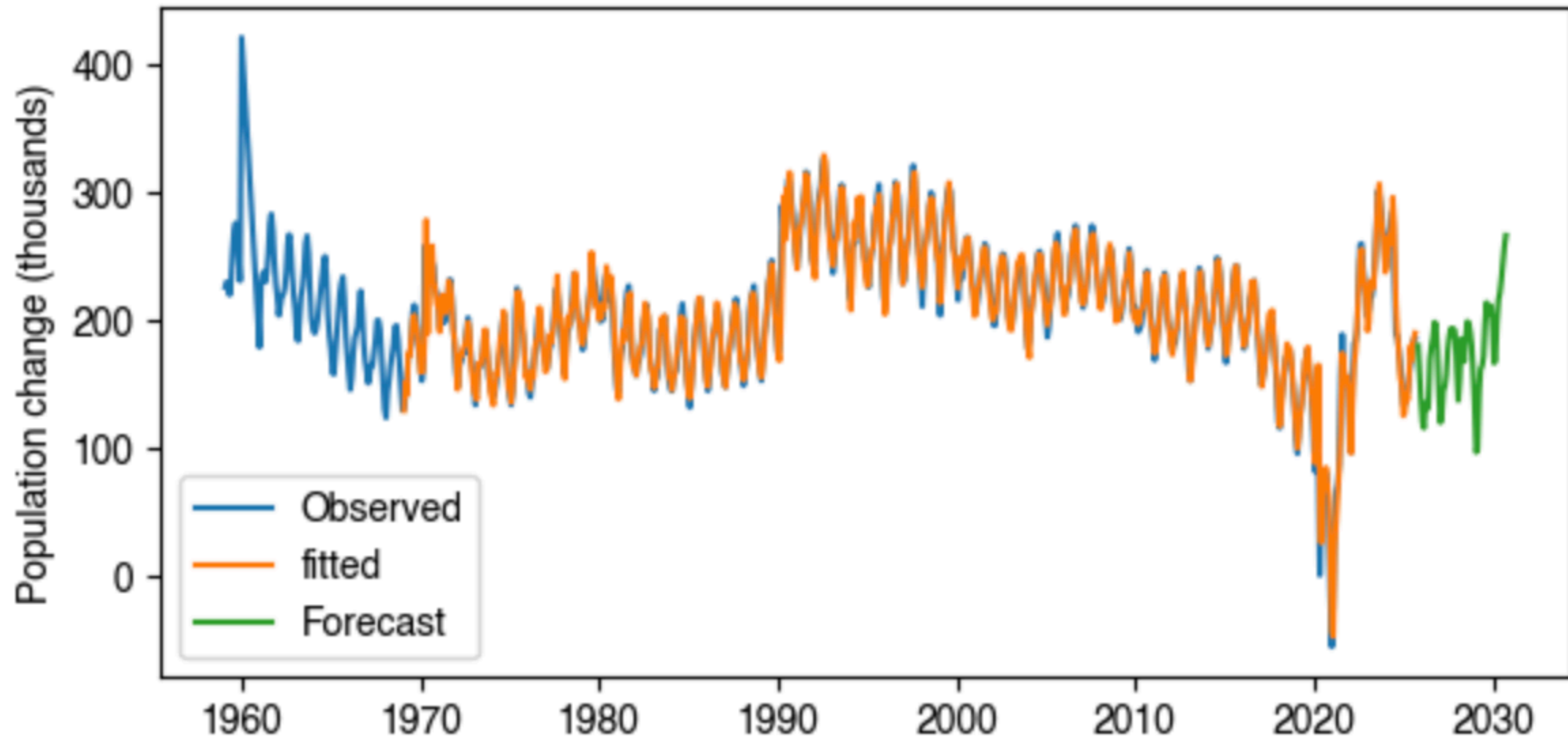
AR, ARIMA, and time-lagged regression

- What is the next number in this series?
 - 1, 1, 2, 3, 5, 8, 13, 21, 34, ...?
- This is the Fibonacci sequence ($y_t = y_{t-1} + y_{t-2}$)
- We cannot just fit a linear regression over time
- Must regress y_t over lagged values y_{t-1} and y_{t-2}
 - Called **AutoRegression (AR)**

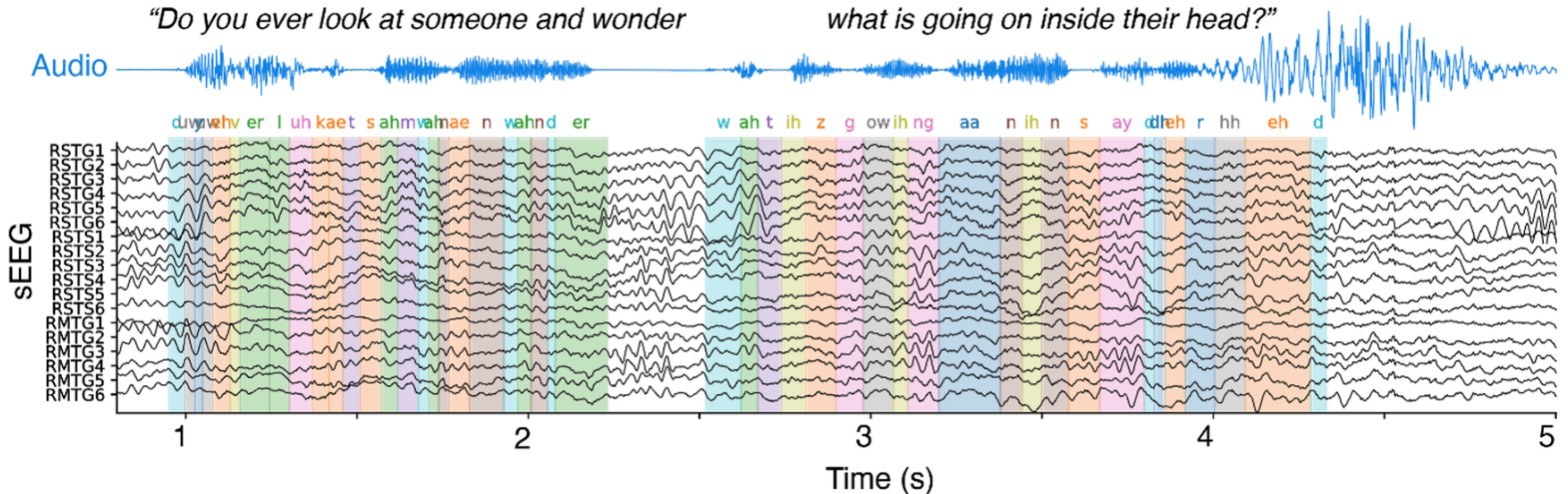
Autoregressive models

- Using prior data from a time series to predict current data (AutoRegression) is the basis of ARIMA models (AutoRegressive Integrated Moving Average Models)
- Do a linear regression of y_t on $x_t = (1, y_{t-1}, \dots, y_{t-p})$ for fixed lag p

Predictions given by AR model incorporating data for previous 10 years

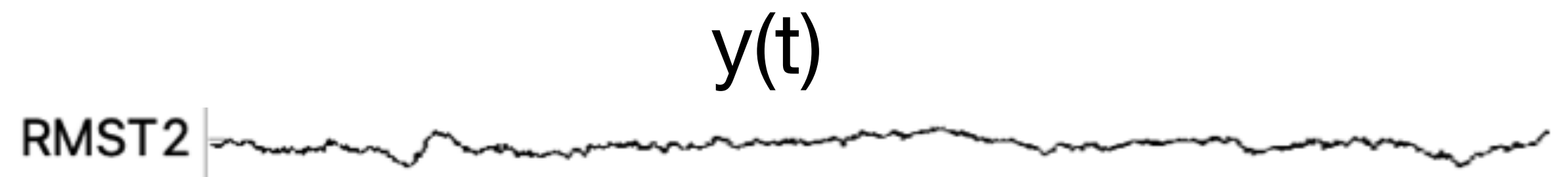


Time-lagged regression models



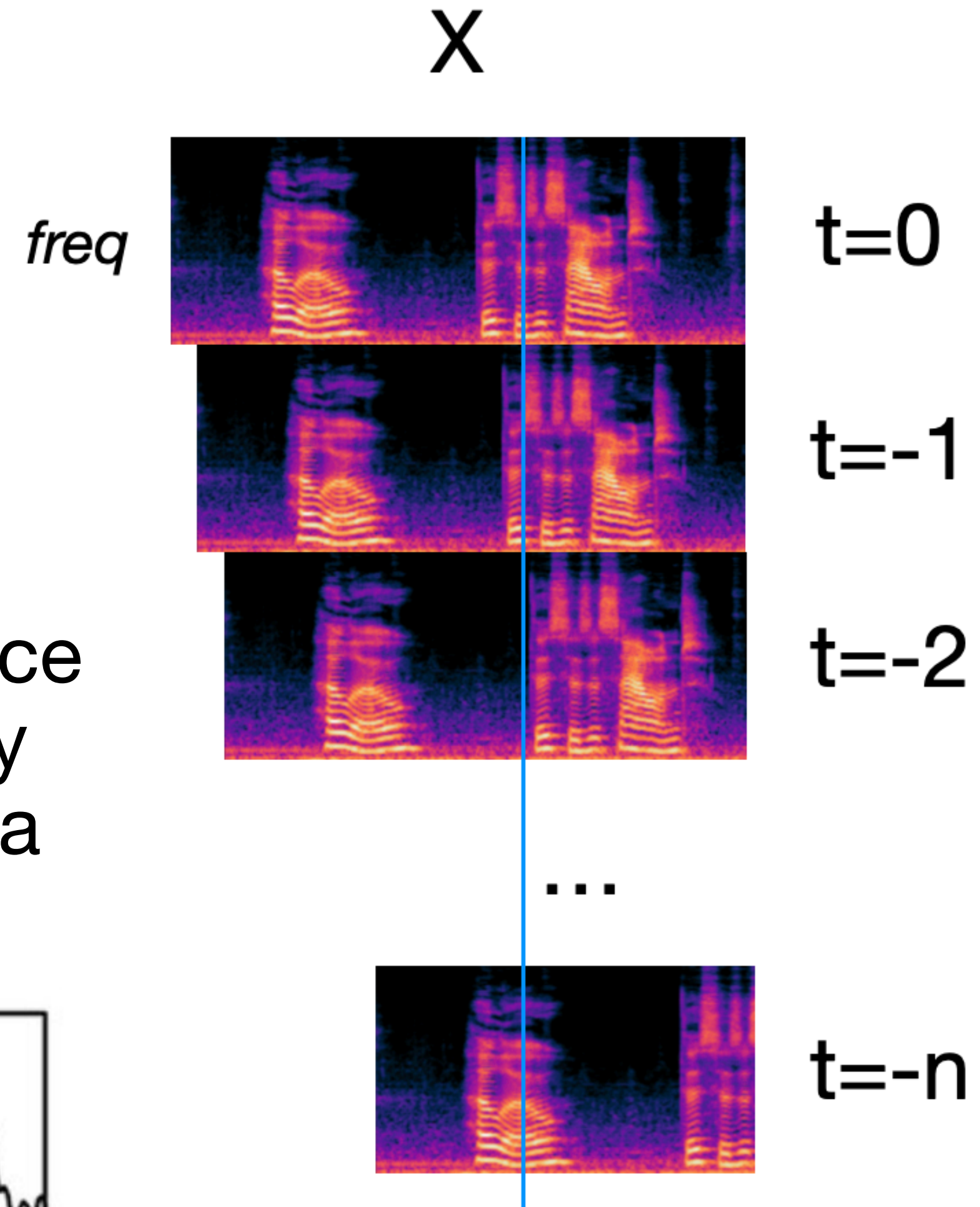
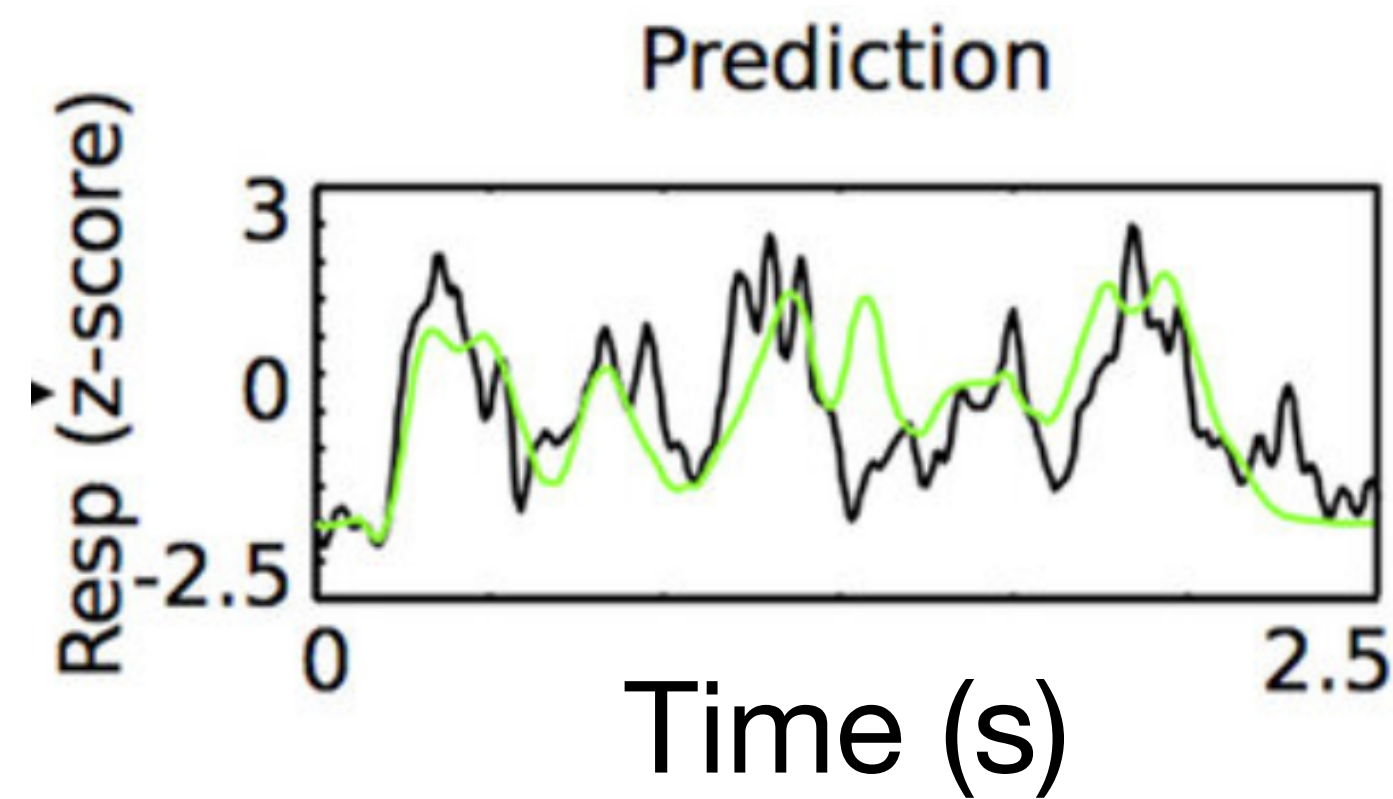
- Brain responses to speech modeled as a function of time-lagged sound features
- We can test different features (sound frequencies, loudness, phonemes)

Time-lagged regression models



$$y(t) = \beta_0 + \sum_f \sum_{\tau} \beta(f, \tau) X(f, t - \tau) + \epsilon$$

These models are used widely in neuroscience to describe what aspects of different sensory stimuli increase or decrease brain activity in a given area



Modern approaches

- We will also learn about **deep neural network** approaches that can be used to predict continuous time series
- Nonlinear, nonparametric approach
- We will discuss **convolutional neural networks (CNNs)**, **recurrent neural networks (RNNs)**, and **self-supervised models**

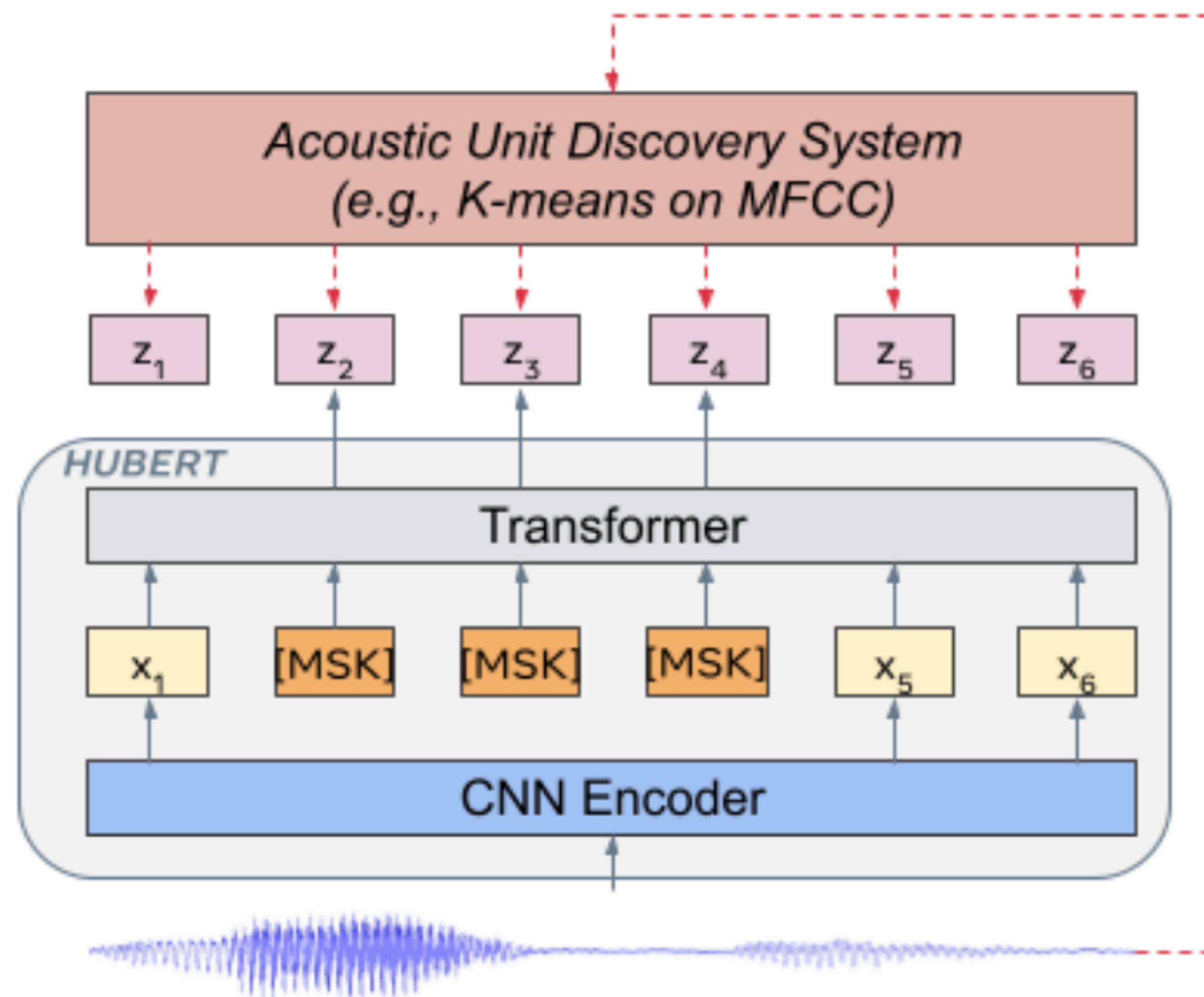


Fig. 1: The HuBERT approach predicts hidden cluster assignments of the masked frames (y_2, y_3, y_4 in the figure) generated by one or more iterations of k-means clustering.

Summary of topics

- Measures of dependence
- Linear regression - simple, multiple, nonlinear, regularization
- Power spectral analysis and time-frequency analysis
- AR, ARIMA, time-lagged regression
- State-space models
- Neural networks - CNNs, RNNs, self-supervised learning (WavLM)

Before next time

- If you haven't done it yet, fill out the pre-semester survey!
- <https://forms.gle/2jLWHGeAvXe9EMcG9>
(or use QR code)
- Next time we will talk about characteristics of time series data

